# Jayesh Khullar

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## Experience

Sep 2023 – Present **Research Assistant, Royal Military College of Canada**

- Ported and configured an autonomous-systems simulation stack to run on an HPC cluster with Slurm and Singularity, enabling large-scale reproducible experiment execution.
- Co-designed a multi-agent drone simulation environment and implemented A* pathfinding to support research into multi-agent coordination and reinforcement learning.
- Contributed to a TMLR submission on temporal representation in recurrent networks, supporting experiments and analysis of SMDP, POMDP, and GSMDP frameworks.
- Supervising a 4th-year Computer Science undergraduate research student.

Aug 2022 – Jun 2026 **Software Engineer & Consultant, OSS Consultants**

- Built an LLM-powered document-review tool with a retrieval-augmented generation pipeline.
- Automated open-source and license-compliance analysis with async Python and REST APIs, including SPDX SBOM generation, across 150+ client software products.
- Met with and presented to end clients to understand their needs, then designed and delivered custom solutions including tailored dashboards, REST APIs, and data migrations.
- Built a custom dashboard letting analysts audit and compare 150+ client projects, saving 200+ hours of manual review per release cycle.
- Migrated 100,000s of records from legacy client databases into a consolidated system, saving 100s of hours and reducing errors.

May 2021 – Aug 2021 **Open Source Developer, BlackBerry**

- Built Python CLI tools for dependency-graph analysis across 150+ Maven products, cutting 100+ analyst hours in license-compliance review.

## Projects

### Deep RL Option Discovery

- Post-hoc analysis method interpreting deep RL agents by clustering hidden-layer activations to surface temporally extended, option-like behaviours.
- Validated across Four-Room gridworld and Atari (Seaquest, Ms. Pac-Man, Montezuma's Revenge, Breakout) via dwell-time distributions, action n-gram statistics, and chi-square validation.
- Built analysis tooling (cluster timelines, spatial heatmaps, duration distributions) across PPO, A2C, DQN, and DDQN agents in PyTorch/SB3.

### LLM Document Review & RAG System

- Local LLM-powered document review using retrieval-augmented generation, vector search, and multi-format document ingestion.
- Evaluated model performance across accuracy, latency, and hardware constraints to drive practical deployment decisions.

## Skills

**Languages:** Python, R, Bash
**AI/ML:** PyTorch, scikit-learn, Stable-Baselines3, clustering & dimensionality reduction
**LLM/RAG:** RAG pipelines, ChromaDB, Ollama, Qwen2.5, nomic-embed-text, local LLM inference, document ingestion
**Data & Infra:** Docker, AWS, REST APIs, MongoDB, MySQL, Git, Linux, Slurm, Singularity, Weights & Biases

## Education

**M.Sc. Computer Science, Queen's University**
GPA 4.30/4.30

**Hon. B.Sc. Statistical ML & Data Mining , University of Toronto**

## Publications & Service

"Detecting temporal abstractions." RLC Continual RL Workshop, 2026.

Peer Reviewer — Reinforcement Learning Conference (RLC), 2026.

"Temporal representation and inter-stimulus interval coding in recurrent networks." Under review, TMLR, 2026.

"Parity Requires Unified Input Dependence and Negative Eigenvalues in SSMs." ICML MOSS Workshop, 2025.